

Nihal Rao

nrao1210@terpmail.umd.edu | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Maryland

Bachelor of Science in Computer Science, Minor in Robotics and Autonomous Systems

- **GPA:** 4.0/4.0

College Park, MD

May 2029

EXPERIENCE

Technical Coordinator & Web Manager

Aug. 2024 – May 2026

Indian Association of Memphis

Memphis, TN

- Directed IT operations and digital infrastructure for a 501(c)(3) non-profit, successfully supporting 40+ annual events with over 5,000+ total attendees
- Engineered promotional web pages and marketing assets, increasing community engagement and driving 1,200+ online event registrations
- Provided technical support for event logistics, audio-visual systems, and guest networking

Computational Physics Researcher

June 2025 – July 2025

TN Governor's School for Computational Physics

Clarksville, TN

- Selected from the top 5% of state applicants for a rigorous 140-hour collegiate program to mathematically model complex multi-variable physical systems
- Developed 10+ complex simulations (3D projectile motion, thermal transfer) using Fortran and Python
- Trained 3 deep learning models using TensorFlow, achieving 92% validation accuracy in predicting non-linear physical system behaviors and kinetic dynamics

PROJECTS

Sentinel (Predictive Maintenance) | *Python, scikit-learn, pandas, Streamlit*

- Built failure-risk and remaining-useful-life models on NASA C-MAPSS simulated engine telemetry using causal time-series features and machine-disjoint training, validation, and test evaluation
- Achieved 0.949 average precision, 84% recall, and 87.5% precision for failure within 30 operating cycles on 100 held-out test engines; reduced capped remaining-life MAE from 38.0 to 12.9 cycles
- Implemented an interactive inference app with historical replay, batch scoring, and simulated maintenance alerts; added six automated tests for leakage prevention, inference, and app behavior

FocusMate (Cognitive Accessibility App) | *React, Python, OpenCV, Flask, Figma*

- Designed and developed a full-stack web application to provide cognitive support for students with ADHD
- Engineered a computer vision backend with Python/OpenCV to analyze webcam feeds at 30 FPS, achieving 90.5% accuracy in facial-marker distraction detection
- Decreased user distraction intervals by an estimated 40% via an automated Flask trigger system that gently blurs on-screen content when a loss of focus is detected
- Built a low-stimulation frontend using React and Figma, iterating on UI components based on A/B testing feedback

InterceptIQ (Shipment Risk Prediction) | *Python, FastAPI, PostgreSQL, RabbitMQ*

- Built a shipment escape-risk model on 18,000 synthetic packages with chronological evaluation; achieved 0.321 average precision versus a 0.153 prevalence baseline on 2,908 held-out packages
- Implemented event-ingestion APIs, source-specific stop detection, and a PostgreSQL transactional outbox with RabbitMQ delivery; verified 50 simulated escape scenarios and duplicate suppression
- Validated late-event retractions, automatic broker outage recovery, and persistence across container recreation; deployed a Streamlit prediction demo and added 12 automated tests

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL (PostgreSQL), HTML/CSS, Fortran

Frameworks: React, Node.js, Express, FastAPI, Flask, Streamlit, OpenCV

Developer Tools: Git, Docker, RabbitMQ, Figma, VS Code, Postman, Jira

Libraries: scikit-learn, TensorFlow, NumPy, pandas, Matplotlib

Certifications: Microsoft Certified: Azure AI Engineer (AI-102), PCEP, PCAP (Python)

Honors: 1st Place in Coding and Programming (FBLA), Top 24 Internationally in Data Science and Analytics (TSA)